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RESEARCH INTERESTS

Low-dimensional topology, in particular knots and links, knot concordance and cobordisms, braids, notions of braid positivity, knotted surfaces, 3- and 4-manifolds, and Heegaard Floer and Khovanov homology.

EMPLOYMENT

12/2025 – 11/2027	Postdoctoral fellow , University of Glasgow , Glasgow, UK. Supported by an SNSF Postdoc.Mobility fellowship. Mentor: Brendan Owens.
10/2023 – 11/2025	Postdoctoral fellow , Max Planck Institute for Mathematics (MPIM), Bonn, Germany. Mentor: Arunima Ray.
07/2019 – 07/2023	Scientific assistant , ETH Zurich , Zurich, Switzerland.
10/2018 – 04/2019	Teaching assistant , Ulm University , Ulm, Germany.
08/2017 – 05/2018	Teaching assistant , Syracuse University , Syracuse, NY, USA.
10/2013 – 07/2017	Teaching assistant , Ulm University , Ulm, Germany.

EDUCATION

07/2019 – 07/2023	Doctor of Science in Mathematics, ETH Zurich , Zurich, Switzerland. Thesis: <i>On notions of braid positivity and knot concordance</i> . Supervisor: Peter Feller. Doctoral examination: 5 June 2023.
08/2016 – 02/2019	Master of Science in Mathematics (1.0 on the German scale, where 1.0 is the highest grade; with distinction), Ulm University , Ulm, Germany. Minor: Computer Science. Thesis: <i>Massey products and linking numbers</i> . Supervisor: Stefan Wewers.
08/2017 – 05/2018	Master of Science in Mathematics (GPA 4.0), Syracuse University , Syracuse, NY, USA (dual-degree exchange programme with Ulm University).
10/2012 – 08/2016	Bachelor of Science in Mathematics (1.5 on the German scale), Ulm University , Ulm, Germany. Minor: Physics. Thesis: <i>Das Existenzproblem von Hurwitz (The Hurwitz existence problem)</i> . Supervisor: Stefan Wewers.
09/2014 – 02/2015	Visiting Semester , ETH Zurich , Zurich, Switzerland.

SELECTED FELLOWSHIPS AND AWARDS

12/2025 – 11/2027	Swiss National Science Foundation Postdoc.Mobility fellowship . Project 230329: Concordance of braid positive knots . CHF 111,600.
08/2017	Fulbright Travel Grant . €2,000.
10/2014 – 09/2017	Talanx Scholarship . €10,800 ($3 \times 12 \times \text{€}300$).
10/2013 – 09/2014	Deutschlandstipendium (Ulm University/Federal Ministry of Education and Research, EY Foundation). €3,600 ($12 \times \text{€}300$).
10/2012 – 09/2013	Deutschlandstipendium (Ulm University/Federal Ministry of Education and Research, Rotary Club Ulm). €3,600 ($12 \times \text{€}300$).

RESEARCH PROGRAMME

Participation in the **AIM Online Research Community 4-Dimensional Topology** organised by Miriam Kuzbary, Maggie Miller, Juanita Pinzón-Cañedo and Hannah Schwartz in Spring 2021.

PREPRINTS

3. **The Conway knot has infinite concordance order**, with C. Donatone, M. Kegel and L. Lewark.
Submitted. [arXiv:2606.30422](#).
2. **On $\mathcal{T}$ -positive links**, with B. Bode.
Submitted. [arXiv:2605.10502](#).
1. **On the detection of knotted spheres by their traces in high dimensions**, with V. Bais, A. Di Prisa, D. Hartman, C.-S. Hsueh, M. Kegel, A. Merz, M. Pencovitch, A. Ray, D. Santoro and L. Wakelin.
Submitted. [arXiv:2511.07251](#).

PUBLISHED AND ACCEPTED PAPERS

8. **Braid positive surgery diagrams**, with M. Kegel.
Glasg. Math. J., published online (2026), pp. 1–6. [arXiv:2512.14589](#).
7. **Non-complex cobordisms between quasipositive knots**, with M. Borodzik.
J. Math. Pures Appl. (9) 207 (2026), Paper No. 103842, 11 pp. [arXiv:2504.04894](#).
6. **Algorithms in 4-manifold topology**, with S. Bastl, R. Burke, R. Chatterjee, S. Dey, A. Durst, S. Friedl, D. Galvin, A. García Rivas, T. Hirsch, C. Hobohm, C.-S. Hsueh, M. Kegel, F. Kern, S. M. S. Lee, C. Löh, N. Manikandan, L. Mousseau, L. Munser, M. Pencovitch, P. Perras, M. Powell, J. P. Quintanilha, L. Schambeck, D. Suchodoll, M. Tancer, A. Thiele, M. Uschold, S. Veselá, M. Weiß and M. von Wunsch-Rolshoven.
Algebr. Geom. Topol. (accepted). [arXiv:2411.08775](#).
5. **Minimal cobordisms between thin and thick torus knots**, with S. Baader, L. Lewark and F. Misev.
Proc. Edinb. Math. Soc. (2) 69 (2026), no. 1, pp. 177–183. [arXiv:2405.13719](#).
4. **3-braid knots with maximal 4-genus**, with S. Baader, L. Lewark and F. Misev.
Trans. Amer. Math. Soc. Ser. B 11 (2024), pp. 600–621. [arXiv:2303.11918](#).
3. **Strongly quasipositive links are concordant to infinitely many strongly quasipositive links**.
Proc. Amer. Math. Soc. (accepted). [arXiv:2210.06612](#).
2. **On the nonorientable four-ball genus of torus knots**, with F. Binns, S. Kang and J. Simone.
Algebr. Geom. Topol. 25 (2025), no. 4, pp. 2209–2251. [arXiv:2109.09187](#).
1. **The upsilon invariant at 1 of 3-braid knots**.
Algebr. Geom. Topol. 23 (2023), no. 8, pp. 3763–3804. [arXiv:2108.03674](#).

Preprint versions of all these articles can be found on [arXiv](#).

CONFERENCE PROCEEDINGS (NOT PEER-REVIEWED)

6. **The Conway knot has infinite concordance order**, with C. Donatone, M. Kegel and L. Lewark.
Oberwolfach Reports 23 (2026), preliminary version, pp. 23–25.
5. **MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation**, with S. Friedl, J. Licata and S. Tillmann.
Oberwolfach Reports 21 (2024), no. 3, pp. 2511–2550.
4. **Obstructing ribbon clasp number one**, with D. Celoria, A. Di Prisa, J. Howie, J. Licata, F. Misev, L. Schambeck and L. Wakelin.
Oberwolfach Reports 21 (2024), no. 3, pp. 2528–2530.
3. **Notions of braid positivity and knot concordance**.
Oberwolfach Reports 21 (2024), no. 3, pp. 2543–2546.
2. **Strongly quasipositive knots are concordant to infinitely many strongly quasipositive knots**.
Oberwolfach Reports 20 (2023), no. 1, pp. 172–174.
1. **Concordance of positive braid knots**.
Oberwolfach Reports 18 (2021), no. 3, pp. 2360–2362.

INVITED TALKS AT CONFERENCES AND WORKSHOPS (16 IN TOTAL)

- 10/2026* [Journées de topologie quantique](#) (quantum topology days), IMJ-PRG, Paris, France.
07/2026 [Young Women in Geometric Topology](#), HCM Bonn, Germany.
07/2026 [Low-dimensional topology](#), MFO Oberwolfach, Germany.
06/2026 [Berlin–Brandenburg Workshop VII: Knot theory and its applications](#), HU Berlin, Germany.
10/2025 [Trisections and related topics](#), CIRM, Marseille, France.
11/2024 [IberoSing International Workshop 2024: Low-dimensional Topology & Singularity Theory](#), UPM Madrid, Spain.
07/2024 [European Congress of Mathematics](#), Mini-symposium *Knot theory and low-dimensional manifolds*, University of Seville, Spain.
04/2024 [33rd NRW Topology Meeting](#), University of Wuppertal, Germany.
04/2024 [Manifolds and groups in Bologna II](#), University of Bologna, Italy.
09/2023 [Leaning into Topology Workshop](#), University of Pisa, Italy.
09/2023 [Swiss Knots](#), University of Regensburg, Germany.
06/2023 Mini workshop on low-dimensional topology, University of Bern, Switzerland.
03/2023 [The low-dimensional workshop](#), Rényi Institute, Budapest, Hungary.
11/2022 [Paroles aux jeunes chercheuses et chercheurs en géométrie et dynamique](#), Université Paris-Saclay, Orsay, France.
09/2022 [DMV Annual Meeting](#), Mini-symposium *Algebra and Low-Dimensional Topology*, FU Berlin, Germany.
06/2022 [Surfaces in 4-manifolds](#), Le Croisic, France.

INVITED TALKS IN RESEARCH SEMINARS (20 IN TOTAL)

- 02/2026 [Geometry & Topology Seminar](#), University of Glasgow, UK.
01/2026 [Geometry Seminar](#), MPI MiS Leipzig, Germany.
10/2025 [Algebra and Geometry Seminar](#), University of Neuchâtel, Switzerland.
07/2025 [Oberseminar Global Analysis and Geometry](#), University of Regensburg, Germany.
06/2025 [Séminaire Géométrie des Espaces Singuliers](#), Université de Lille, France.
05/2025 [Geometry Seminar](#), ETH Zurich, Switzerland.
12/2024 [Oberseminar Topologie](#), University of Bonn, Germany.
09/2024 [Séminaire de Géométrie, Algèbre, Dynamique et Topologie](#), Université de Bourgogne, Dijon, France.
05/2024 [Séminaire de Géométrie et Topologie de Marseille](#), Aix-Marseille Université, France.
05/2024 [Oberseminar Topologie](#), University of Münster, Germany.
04/2024 [Geometry & Topology Seminar](#), University of Glasgow, UK.
03/2024 [MPIM Oberseminar](#), MPIM Bonn, Germany.
11/2023 [MPIM Topology Seminar](#), MPIM Bonn, Germany, short talk.
05/2023 [Low-dimensional topology seminar](#), MPIM Bonn, Germany.
12/2022 [Geometry & Topology Seminar](#), CIRGET, UQAM, Montréal, Canada, online.
06/2022 [LKS-Seminar](#), University of Regensburg, Germany.
05/2022 [Geometry & Topology Seminar](#), Caltech, Pasadena, CA, USA, online.
05/2022 [Oberseminar Geometrie](#), University of Fribourg, Switzerland.
02/2022 [Knot Online Seminar \(K-OS\)](#), online.
04/2021 [AIM Online Research Community 4-Dimensional Topology](#), lightning talk, online.

INVITED TALKS AT GRADUATE COLLOQUIA (7 IN TOTAL)

- 12/2023 [Pitt AWM Student Seminar](#), University of Pittsburgh, PA, USA, online.
10/2023 [Geometry Graduate Colloquium](#), ETH Zurich, Switzerland.
10/2022 [Bern–Fribourg Graduate Seminar](#), University of Bern, Switzerland.
05/2022 [Baby Geometri Seminar](#) (Pisa junior geometry seminar), University of Pisa, Italy.
04/2022 [Bernoullis Tafelrunde](#), University of Basel, Switzerland.
11/2021 [Zurich Graduate Colloquium](#), ETH Zurich and University of Zurich, Switzerland.
04/2021 [Geometric Topology Grad and Postdoc Seminar](#) (GT GAPS), online.

* means planned.

CONTRIBUTED TALKS AT CONFERENCES AND WORKSHOPS (7 IN TOTAL)

- 09/2024 [MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation](#), MFO Oberwolfach, Germany, short talk.
- 02/2024 [BTW4: Knot Floer Homology](#), Besse, France.
- 02/2023 [Winter Braids XII](#), University of Tours, France.
- 01/2023 [Low-dimensional topology](#), MFO Oberwolfach, Germany.
- 05/2022 [Swiss Mathematical Society Doctoral Day 2022](#), University of Fribourg, Switzerland.
- 04/2022 [BTW2: Sliceness, exotic pairs, and quantum invariants](#), Aussois, France.
- 09/2021 [Invariants and Structures in Low-Dimensional Topology](#), MFO Oberwolfach, Germany, short talk.

OTHER CONFERENCES AND WORKSHOPS ATTENDED (24 IN TOTAL)

- 06/2026 [Geometry and Topology in Low Dimensions: Lasting Trends and Emerging Directions](#), Hungarian Academy of Sciences, Budapest, Hungary.
- 12/2025 [Workshop on surfaces in 4-manifolds](#), University of Regensburg, Germany.
- 11/2025 [36th NRW Topology Meeting](#), University of Bonn, Germany.
- 07/2025 [Categorification in Low Dimensional Topology](#), Ruhr University Bochum, Germany.
- 06/2025 [Swiss Knots](#), University of Geneva, Switzerland.
- 09/2024 [Workshop on 4-manifolds and algorithms](#), University of Regensburg, Germany.
- 02/2024 [Winter Braids XIII](#), University of Montpellier, France.
- 09/2023 [GRAZP: Groups and Rigidity Around the Zimmer Program](#), Ventotene, Italy.
- 07/2023 [Gauge Theory and Topology: in Celebration of Peter Kronheimer's 60th Birthday](#), University of Oxford, UK.
- 01/2023 [Winter school on singularities and low-dimensional topology](#), Rényi Institute, Budapest, Hungary.
- 08/2022 [Frontiers in Geometry and Topology Summer School](#), ICTP, Trieste, Italy.
- 04/2022 [Braids in Low-Dimensional Topology](#), ICERM, Providence, RI, USA.
- 08/2021 [Perspectives on quantum link homology theories](#), University of Regensburg, Germany.
- 07/2021 [Tech Topology Summer School](#), Georgia Institute of Technology, Atlanta, GA, USA, online.
- 07/2021 [Swiss Knots](#), University of Fribourg, Switzerland.
- 06/2021 [NCNGT: Nearly Carbon Neutral Geometric Topology Conference](#), online.
- 06/2020 [NCNGT: Nearly Carbon Neutral Geometric Topology Conference](#), online.
- 02/2020 [Winter Braids X](#), University of Pisa, Italy.
- 10/2019 [Regional workshop on foliations and 3-manifolds](#), University of Regensburg, Germany.
- 10/2019 [Low-dimensional topology workshop](#), University of Regensburg, Germany.
- 09/2019 [Workshop on 4-manifolds](#), MPIM Bonn, Germany.
- 09/2019 [Of coarse! Quasi-isometries and groups: rigidity and classification](#), Ventotene, Italy.
- 07/2019 [Swiss Knots](#), ETH Zurich, Switzerland.
- 10/2017 [27th annual Route 81 conference on commutative algebra and algebraic geometry](#), Cornell University, Ithaca, NY, USA.

SHORT RESEARCH STAYS WITHOUT TALK, CONFERENCE OR WORKSHOP

- 07/2025 three days, HU Berlin, Germany.
- 08/2024 one week, ETH Zurich, Switzerland.
- 03/2024 three days, ETH Zurich, Switzerland.
- 04/2022 one week, ICERM, Providence, RI, USA.
- 09/2020 three days, University of Regensburg, Germany.

SUPERVISION

- 02/2025 – 10/2025 **Master's thesis** by Chiara Donatone at the University of Milan (during her Erasmus stay at the University of Bonn). Co-supervised by Diego Matessi.
Thesis title: *On RGB links and knot traces*.
Continues as a PhD student at the University of Glasgow (since September 2026).
- 06/2024 – 06/2025 **Master's thesis** by Frieda Kern at the University of Bonn.
Co-supervised by Peter Teichner.
Thesis title: *Linking number obstructions to satellite homomorphisms*.
Continues as a [PhD student](#) at the University of Glasgow (since September 2025).

TEACHING

- Lecturer/instructor
- Winter 2024/2025 **Topics in 3-manifold topology** (Master in Mathematics), University of Bonn.
- Winter 2018/2019 **Linear Algebra review course** (Bachelor in Computer Science and Engineering), Ulm University.
- Course organiser at ETH Zurich
- Spring 2022 **Analysis II** (Bachelor in Civil, Geospatial or Environmental Engineering).
- Autumn 2021 **Analysis I** (Bachelor in Civil, Geospatial or Environmental Engineering).
- Spring 2021 **Topologie** (Bachelor in Mathematics).
- Autumn 2020 **Lineare Algebra I** (Bachelor in Mathematics, Physics or Interdisciplinary Sciences).
- Spring 2020 **Algebraic Topology II** (Bachelor and Master in Mathematics).
- Teaching assistant
- Autumn 2019 **Complex Analysis** (Bachelor in Mathematics, Physics or Interdisciplinary Sciences), ETH Zurich.
- Spring 2018 **Calculus III**, Syracuse University.
- Autumn 2017 **Calculus III**, Syracuse University.
- Winter 2013/2014 – Winter 2018/2019 Student teaching assistant for **Analysis I–III** and **Linear Algebra I–II**, Ulm University.

PEDAGOGICAL TRAINING

- 09/2024 – 01/2025 Participation in the **Teaching Mentoring Programme**, Hausdorff School for Mathematics in Bonn. Mentors: Viktoriya Ozornova and Arunima Ray (both MPIM at the time).
- 09/2021 Certificate **Learning to Teach**.
Programme for doctoral teaching assistants, ETH Zurich.
Teaching project: *How to teach writing proofs: classroom assessment techniques in mathematics classes*.

OUTREACH AND COMMUNITY-BUILDING ACTIVITIES

- 07/2026 Invited panellist at **Young Women in Geometric Topology**, HCM Bonn.
- 03/2026 Co-organiser of **GROW@Dresden 2026** (Graduate Research Opportunities for Women at Dresden 2026), TU Dresden and MPI-CBG, 26–27 March. Joint with Anita Behme, Türkü Özlüm Çelik, Heather Harrington, Oana Padurariu and Andreas Thom.
- 03/2025 Invited panellist at **GROW@Frankfurt 2025**, Goethe University Frankfurt.
- 12/2024 Invited speaker at **Tea Time with Women in Mathematics**, University of Bonn.
- 04/2024 Volunteer at the mentoring lunch at **GROW@Bonn 2024**, MPIM Bonn.
- 09/2021 – 07/2023 Co-organiser of **get-togethers for women in mathematics** at ETH Zurich and the University of Zurich. Funded by the Zurich Graduate School in Mathematics.
- 03/2023 *Work as a PhD Student*, invited short talk, **MindPhair 2023**, ETH Zurich (Job fair for mathematicians, physicists and computational scientists).
- 03/2022 *Work as a PhD Student*, invited short talk, **MindPhair 2022**, ETH Zurich.

CONFERENCE ORGANISATION

- 12/2026* **Workshop on Knotted Surfaces**, University of Glasgow, 7–11 December.
Joint with Simeon Hellsten, Frieda Kern and Mark Powell.
- 09/2024 **MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation**, MFO Oberwolfach, 22–27 September. Joint with Stefan Friedl, Joan Licata and Stephan Tillmann.

SEMINAR ORGANISATION

- 09/2026 – **Geometry & Topology Seminar**, University of Glasgow.
Joint with Livio Ferretti.
- 02/2026 – 05/2026 **arXiv seminar**, University of Glasgow. Joint with Livio Ferretti.
- 10/2023 – 11/2025 **Low-dimensional topology seminar**, MPIM Bonn.
Joint variously with Xujia Chen, Daniel Galvin, Daniel Hartman, Roberto Ladu, Arunima Ray, Léo Schelstraete, Isaac Sundberg, Peter Teichner, Laura Wakelin and Hugo Zhou.
- 09/2020 – 07/2023 **Geometry Graduate Colloquium**, ETH Zurich.
Funded by the Zurich Graduate School in Mathematics.
Joint variously with Raphael Appenzeller, Emilio Corso, Luca De Rosa, Xenia Flamm, Francesco Fournier Facio, Martina Jørgensen and Yannick Krifka.

REFEREEING AND REVIEWING

- 2024 – **Refereeing** and providing **quick opinions** for *Algebraic & Geometric Topology*, *Bulletin of the London Mathematical Society*, *Pacific Journal of Mathematics* and *Quantum Topology*.
- 2024 – **Reviewer** for *zbMATH Open*.

OTHER ACADEMIC SERVICES

- 09/2024 Member of the Postdoc Representative Selection Committee, MPIM Bonn.
- 08/2021 – 07/2023 Elected member of the Nachdiplom Lectures Committee at the Department of Mathematics, ETH Zurich.
- 05/2020 – 05/2023 Member of the Board of **VMM** (Association of the Mid-level Academic Faculty at the Department of Mathematics), ETH Zurich.
- 10/2022 Co-organiser of the first D-MATH (Post)Doctoral Welcome Day, ETH Zurich.
- 05/2021 – 05/2022 Mid-level faculty representative in the Department Conference, ETH Zurich.
- 10/2016 – 07/2017 Member of the Mathematics Student Council (Fachschaft Mathematik) at the Department of Mathematics and Economics, Ulm University.

REFERENCES

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| Peter Feller , University of Neuchâtel, | peter.feller@unine.ch. |
| Matthew Hedden , Michigan State University, | mhedden@math.msu.edu. |
| Lukas Lewark , ETH Zurich, | lukas.lewark@math.ethz.ch. |
| Brendan Owens , University of Glasgow, | Brendan.Owens@glasgow.ac.uk. |
| Viktoriya Ozornova , MPIM Bonn (Teaching), | viktoriya.ozornova@mpim-bonn.mpg.de. |
| Arunima Ray , University of Melbourne, | aru.ray@unimelb.edu.au. |